Hao-Yuan He (何浩源)

+86 18851768896 | Nanjing, China | alkane0050@gmail.com | hao-yuan-he.github.io

Education

School of Computer Science, Nanjing Tech University 2017 – 2021
B.Eng. in Computer Science.

School of Artificial Intelligence, Nanjing University 2021 – 2024
M.Eng. in Computer Science, LAMDA Group. Advised by Prof. Ming Li.

School of Artificial Intelligence, Nanjing University 2024 – Now
Ph.D. in Computer Science, LAMDA Group. Advised by Prof. Ming Li.

Work Experience

Research Intern, AI Lab, Tencent 2023.05 – 2023.07
Worked on game AI for Honor of Kings (王者荣耀), migrating the training pipeline from TensorFlow 1.12 to PyTorch and developing novel algorithms for fine-grained, competition-level agents. *Shenzhen, China*

Research

I study reliable neuro-symbolic AI under weak supervision and structured reasoning. Selected publications:

Learning-Augmented Smooth Integer Programs with PAC-Learnable Oracles Submitted to ICML'2026
Developed a learning-augmented framework for NP-hard optimization problems such as MAX-CUT and MAX-k-SAT. The method uses predictions to build a tractable surrogate objective, remains robust to prediction errors, extends approximation guarantees from dense to near-dense settings, and proves the oracle can be learned efficiently from data.

A Learnability Analysis on Neuro-Symbolic Learning NeurIPS'2025 (Spotlight)
Established a characterization of when neuro-symbolic tasks are truly learnable by linking them to their underlying constraint structure.

Ambiguity-Aware Abductive Learning ICML'2024
Proposed an ambiguity-aware approach for abductive learning when logical reasoning returns multiple plausible supervision candidates. Instead of committing to one candidate too early, the method models uncertainty over candidates, yielding more reliable supervision and stronger performance.

Academic Service

Conference Reviewer / PC Member 2023 – Now
Reviewed for ACML, IJCLR, ICLR, ICML, NeurIPS, and AAAI.

Journal Reviewer 2023 – Now
Reviewed for Knowledge and Information Systems.

Awards

Outstanding Graduate Student 2023, 2024
Nanjing University

Winner 2022
International Algorithm Case Competition, Track of Data Selection and Label Correction

First Prize of Academic Scholarship 2022, 2023
Top 10%, Nanjing University

National Encouragement Scholarship 2019
Jiangsu Education Department

First Prize, Chinese Mathematics Competitions 2018, 2020
Chinese Mathematical Society

Publications

[1] H.-Y. He and M. Li, “A Learnability Analysis on Neuro-Symbolic Learning,” in *NeurIPS*, 2025.

[2] H.-Y. He, Y. Liu, R.-B. Liu, Z. Xie, and M. Li, “Probabilistic Instance Dependent Label Refinement for Noisy Label Learning,” *Machine Learning*, 2025.

[3] H.-Y. He, H. Sun, Z. Xie, and M. Li, “Ambiguity-Aware Abductive Learning,” in *ICML*, 2024.

[4] H.-Y. He, W.-Z. Dai, and M. Li, “Reduced Implication-bias Logic Loss for Neuro-Symbolic Learning,” *Machine Learning*, 2024.

[5] Z. Xie, Y. Liu, H.-Y. He, M. Li, and Z.-H. Zhou, “Weakly Supervised AUC Optimization: A Unified Partial AUC Approach,” *TPAMI*, 2024.

[6] H. Sun, Z. Xie, H.-Y. He, and M. Li, “Mitigating Negative Transfer via Reducing Environmental Disagreement,” *TPAMI*, 2026.

[7] H.-Y. He and M. Li, “Learning-Augmented Smooth Integer Programs with PAC-Learnable Oracles,” *Submitted to ICML'2026*.